

Selina Wang

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Education

Harvard College

Cambridge, MA

Bachelor of Arts – Computer Science joint with Statistics, GPA: 3.90

2025 – 2029

Relevant Courses: Linear Algebra & Vector Calculus, Systems Programming, Probability, Statistical Inference, Algorithms and Limitations | **Selected:** Point72 Academy Spring Sessions

Work Experience

Amazon Web Services – Signer Team

Seattle, WA

Software Engineer Intern

May 2026 – Aug 2026

- Shipped **Profile Aliases for AWS Signer**, eliminating consumer-side policy updates on key rotation for launch partners SSM, EKS, and Amazon Linux across **634+ signing profiles** and **20+ regions** on a service handling **200+ TPS** (load-tested to 1K TPS)
- Engineered **DynamoDB conditional writes** with 3-retry UUID collision mitigation and cross-account Auth Lambda resolution to preserve IAM permission models with **zero public API breaking changes**

Amazon – Traffic Engineering Team

Manhattan, NY

Junior Software Engineer

Feb 2024 – Aug 2025

- Gave leadership **first-ever automated visibility** into an **S-Team mandated** security migration saving **\$1.7M/year** by building an ETL pipeline and compliance dashboard tracking AAA-Symmetric deprecation across **169K+ service relationships** and **35K+ applications** (217B monthly requests)
- Replaced zero prior visibility with hourly-updated time-series metrics using **EventBridge, Lambda, Redshift Federated Queries, and Harmony**, enabling data-driven milestone decisions for a deprecation affecting all Amazon SDO services

Research Experience

Carnegie Mellon University

Remote

Research Assistant – Professor Franceska Xhakaj

June 2024 – Sep 2024

- Engineered an **interactive Unity VR environment** in C# simulating trauma triggers with event-driven scenarios; integrated **OpenAI API** for adaptive real-time conversational support during therapeutic sessions

Leadership & Activities

Harvard Tech for Social Good

Cambridge, MA

Software Engineer

Sep 2025 – Present

- Owned full-stack delivery of **inventory management features** for Breaktime (homeless services platform): **real-time barcode scanner** (USB/camera via ZXing), **role-based CRUD flows** with Zod and react-hook-form, and **analytics dashboards** backed by PostgreSQL RPC functions and Supabase Realtime

Harvard Undergraduate Machine Intelligence Community

Cambridge, MA

Machine Learning Researcher

Sep 2025 – Present

- Cut content production time **from hours to under 5 minutes** by building a **CrewAI multi-agent pipeline** (curriculum design, script writing, and review agents) that converts PDFs into narrated videos, serving **30+ students** with **10+ videos** produced

Harvard AI Safety Student Team (AISST)

Cambridge, MA

Introductory Technical AI Safety Fellow

May 2026 – Present

- Attended meetings covering transformer training, alignment failure modes, mechanistic interpretability, and safety evaluations on frontier models

Projects

benchcheck – LLM Benchmark Contamination Detector:

- Built a **4-signal memorization detector** (n-gram overlap, verbatim completion, perturbation gap, shuffle sensitivity) that quantifies how much of an LLM's benchmark score reflects training data recall vs. genuine capability, without needing access to training data
- Achieved **0% false positives** on clean benchmarks and **91% precision** at 30% contamination; applied to **GPT-2 and Qwen2.5-3B** across 4 public science benchmarks, finding 7–19% contamination rates with consistent rank-order agreement between models
- Validated via calibration experiments on synthetic benchmarks with known contamination rates across **0%, 10%, 30%, and 50%** contamination levels

Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Bash, Java, Swift, HTML/CSS

Frameworks & Libraries: React, Node.js, pandas, NumPy, scikit-learn, PySpark, CrewAI

Tools & Platforms: Docker, Git, AWS (CDK, Lambda, DynamoDB, Redshift), PostgreSQL, Supabase, Firebase, Unity, Linux

Concepts: REST APIs, ETL Pipelines, Serverless Architecture, CI/CD, Machine Learning, LLMs, Multi-Agent Systems